Day – 8 Statistics

Statistics For Machine Learning:

Statistics for Machine Learning is the study of collecting, analyzing and interpreting data to help build better machine learning models. It provides the mathematical foundation to understand data patterns, make predictions and evaluate model performance.

- It helps in understanding data distribution, variability and selecting the most useful features.
- It is used to validate model results and make decisions under uncertainty using hypothesis tests, confidence intervals and Bayesian methods.

Type of Statistics: -

1. Descriptive Statistics
2. Inferential Statistics

1. Descriptive Statistics: - it consists of organizing summarizing and Visualizing data.

	Median	**Mode**
Mean is the sum of all values divided by the total number of values. Mean μ = Sum of Values \ Number of Values	Median is the middle value of sorted data. For odd number of data points: Median= $\left(\frac{n+1}{2}\right)^{th}$ For even number of data points: Median = Average of $\left(\frac{n}{2}\right)^{th}$ value and its next value	Mode is the most frequently occurring value in the dataset.

Inferential Statistics:

Inferential statistics involve making predictions or inferences about a population based on a sample of data.

Measures of inferential statistics are:

T-test

Z-test

CHI Square Test

Anova test

Hypothesis testing

P-Value

Significance value

❖Histograms and Skewness: -

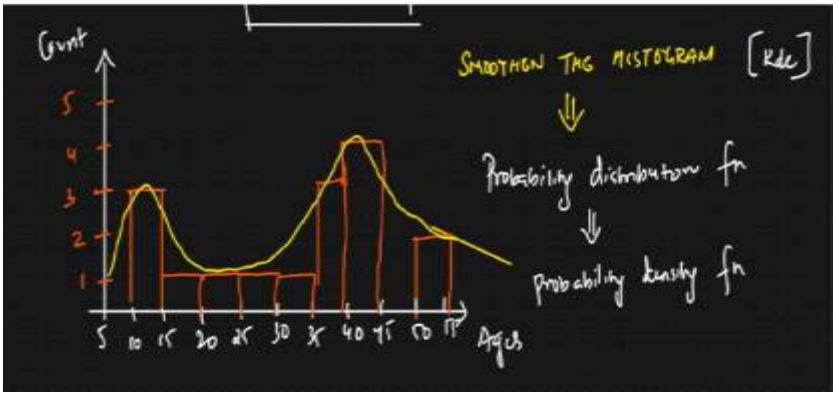
Histogram: -

Ages= {10,12,14,18,24,30,35,36,37,40,41,42,43,50,51}

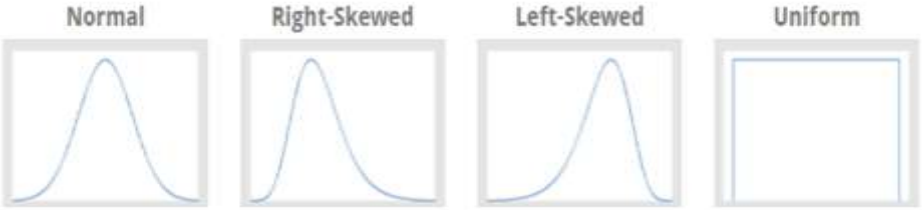
Bins, Bin size

No. of Bins=50/5=10

Bin size=5



Skewness: - Skewness can be defined as a statistical measure that describes the lack of symmetry or asymmetry in the probability distribution of a dataset. It quantifies the degree to which the data deviates from a perfectly symmetrical distribution, such as a normal (bell-shaped) distribution. Skewness is a valuable statistical term because it provides insight into the shape and nature of a dataset's distribution.



❖ sampling Techniques: -



A. Simple random sampling:-

Example: Simple random sampling:- You want to select a simple random sample of 1000 employees of a social media marketing company. You assign a number to every employee in the company database from 1 to 1000, and use a random number generator to select 100 numbers.

B. Stratified sampling:-

Stratified sampling involves dividing the population into subpopulations that may differ in important ways. It allows you draw more precise conclusions by ensuring that every subgroup is properly represented in the sample. To use this sampling method, you divide the population into subgroups (called strata) based on the relevant characteristic (e.g., gender identity, age range, income bracket, job role).

C. Systematic sampling:-

Systematic sampling is similar to simple random sampling, but it is usually slightly easier to conduct. Every member of the population is listed with a number, but instead of randomly generating numbers, individuals are chosen at regular intervals.

Example: Systematic sampling: - All employees of the company are listed in alphabetical order. From the first 10 numbers, you randomly select a starting point: number 6. From number 6 onwards, every 10th person on the list is selected (6, 16, 26, 36, and so on), and you end up with a sample of 100 people.

D. Convenience sampling:-

A convenience sample simply includes the individuals who happen to be most accessible to the researcher.This is an easy and inexpensive way to gather initial data, but there is no way to tell if the sample is representative of the population, so it can't produce generalizable results. Convenience samples are at risk for both sampling bias and selection bias.

Example: Convenience sampling: - You are researching opinions about student support services in your university, so after each of your classes, you ask your fellow students to complete a survey on the topic. This is a convenient way to gather data, but as you only surveyed students taking the same classes as you at the same level,the sample is not representative of all the students at your university.

E. Purposive sampling:-

This type of sampling, also known as judgement sampling, involves the researcher using their expertise to select a sample that is most useful to the purposes of the research. It is often used in qualitative research, where the researcher wants to gain detailed knowledge about a specific phenomenon rather than make statistical inferences, or where the population is very small and specific. An effective purposive sample must have clear criteria and rationale for inclusion. Always make sure to describe your inclusion and exclusion criteria and beware of observer bias affecting your arguments.

Example: Purposive sampling:- You want to know more about the opinions and experiences of disabled students at your university, so you purposefully select a number of students with different support needs in order to gather a varied range of data on their experiences with student services.

F. Cluster sampling:-

Cluster sampling also involves dividing the population into subgroups, but each subgroup should have similar characteristics to the whole sample. Instead of sampling individuals from each subgroup, you randomly select entire subgroups. If it is practically possible, you might include every individual from each sampled cluster. If the clusters themselves are large, you can also sample individuals from within each cluster using one of the techniques above. This is called multistage sampling.

This method is good for dealing with large and dispersed populations, but there is more risk of error in the sample, as there could be substantial differences between clusters. It's difficult to guarantee that the sampled clusters are really representative of the whole population.

Example: Cluster sampling: - The company has offices in 10 cities across the country (all with roughly the same number of employees in similar roles). You don't have the capacity to travel to every office to collect your data, so you use random sampling to select 3 offices – these are your clusters.

❖ Covariance and Correlation: -

• Covariance is a statistical term that refers to a systematic relationship between two random variables in which a change in the other reflects a change in one variable.

• The covariance value can range from -∞ to +∞, with a negative value indicating a negative relationship and a positive value indicating a positive relationship.

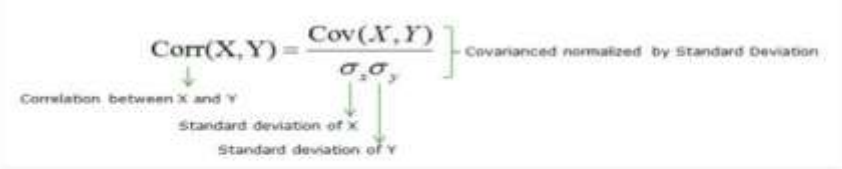
• The greater this number, the more reliant the relationship. Positive covariance denotes a direct relationship and is represented by a positive number.

• A negative number, on the other hand, denotes negative covariance, which indicates an inverse relationship between the two variables. Covariance is great for defining the type of relationship, but it's terrible for interpreting the magnitude.

• Positive: An increase in one of the variables results in an increase in the other.

• Negative: The variables are in opposite directions.

• Zero: Then, no relationship exists.



Covariance explains the joint variability of the variables.

$$\text{Cov}(X,Y) = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{N-1}$$

Where

x_i = Data value of x

y_i = Data value of y

$\bar{x}$ = Mean of x

$\bar{y}$ = Mean of y

N = Number of data values

Types of Probability Distributions

Two major kinds of distributions based on the type of likely values for the variables are,

1. Discrete Distributions

2. Continuous Distributions

1. The probability distribution function / probability function has ambiguous definition. They may be referred to:

• Probability density function (PDF)

• Cumulative distribution function (CDF)

• or probability mass function (PMF)

2. But what confirm is:

• Discrete case: Probability Mass Function (PMF)

• Continuous case: Probability Density Function (PDF)

• Both cases: Cumulative distribution function (CDF)

3. Probability at certain x value, P(X=x) can be directly obtained in:

• PMF for discrete case

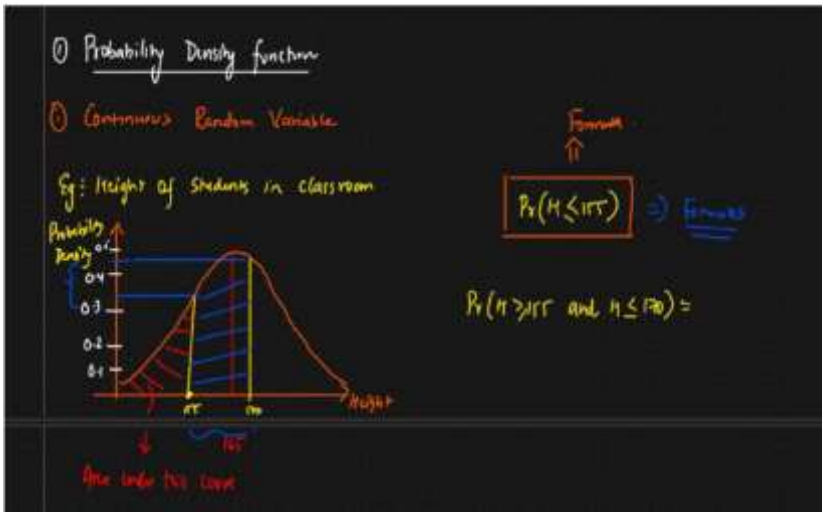
• PDF for continuous case

4. Probability for values less than x, P(X<x) or Probability for values within a range from a to b, P(a<X<b) can be directly obtained in:

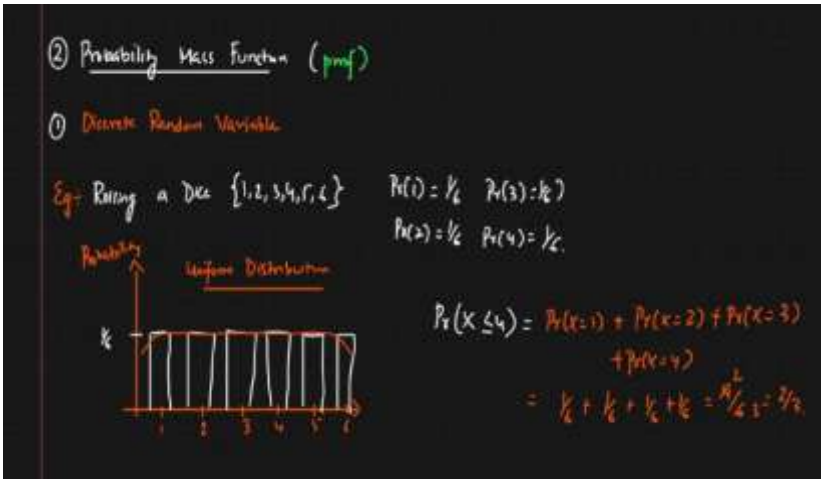
• CDF for both discrete / continuous case

5. Distribution function is referred to CDF or Cumulative Frequency Function.

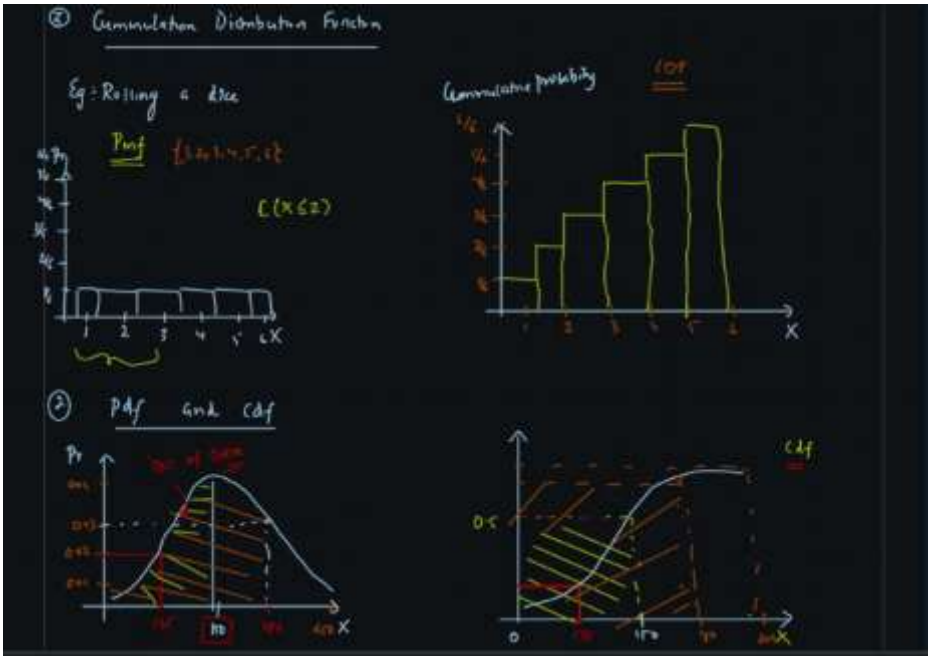
A. Probability Density Function (PDF): - It is a statistical term that describes the probability distribution of a continuous random variable.The probability associate with a single value is always Zero. Below is the formula for PDF.



B. Probability Mass Function (PMF):- It is a statistical term that describes the probability distribution of a discrete random variable.



C. Cumulative Distribution Function (CDF):- It is another method to describe the distribution of a random variable (either continuous or discrete).

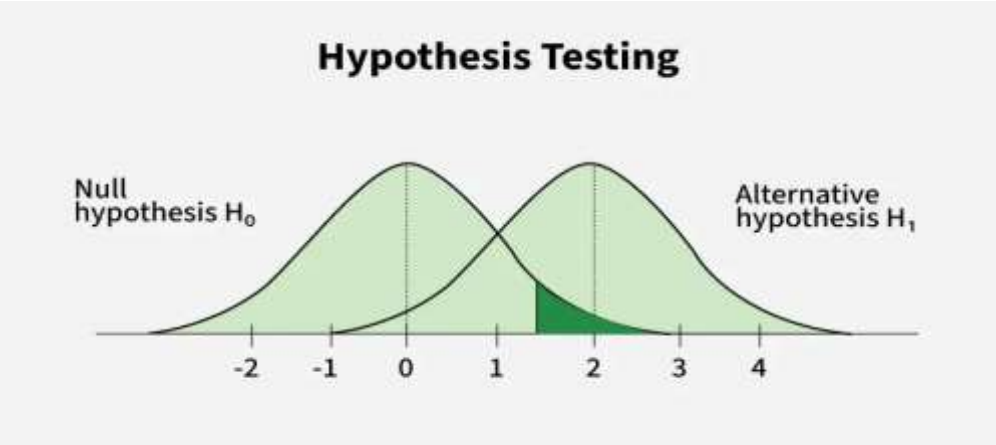


Standardization and Normalization:

S.NO.	Normalization	Standardization
1.	Minimum and maximum value of features are used for scaling	Mean and standard deviation is used for scaling.
2.	It is used when features are of different scales.	It is used when we want to ensure zero mean and unit standard deviation.
3.	Scales values between [0, 1] or [-1, 1].	It is not bounded to a certain range.
4.	It is really affected by outliers.	It is much less affected by outliers.
5.	Scikit-Learn provides a transformer called MinMaxScaler for Normalization.	Scikit-Learn provides a transformer called StandardScaler for standardization.
6.	This transformation squishes the n-dimensional data into an n-dimensional unit hypercube.	It translates the data to the mean vector of original data to the origin and squishes or expands.
7.	It is useful when we don't know about the distribution	It is useful when the feature distribution is Normal or Gaussian.
8.	It is a often called as Scaling Normalization	It is a often called as Z-Score Normalization.

Hypothesis Testing

Hypothesis testing compares two opposite ideas about a group of people or things and uses data from a small part of that group (a sample) to decide which idea is more likely true. We collect and study the sample data to check if the claim is correct.



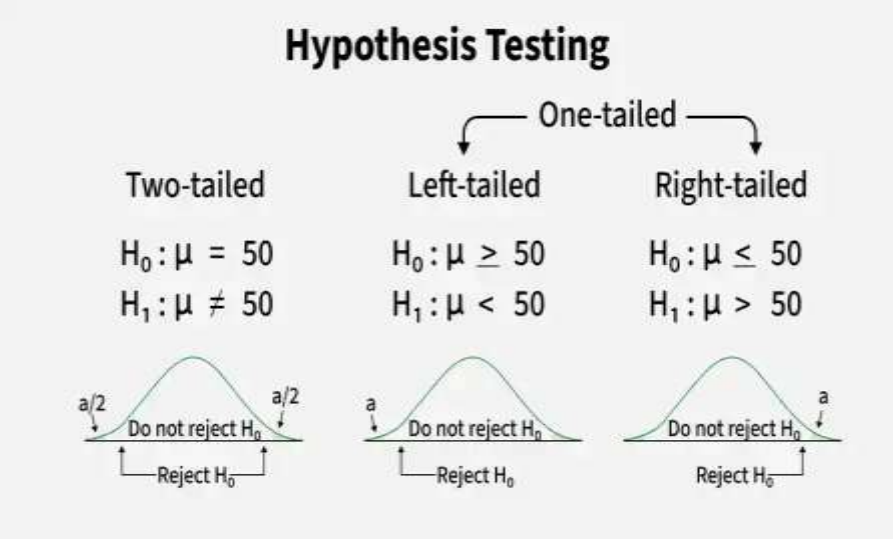
Defining Hypotheses

- **Null Hypothesis (H_0):** The starting assumption. For example, "The average visits are 50."
- **Alternative Hypothesis (H_1):** The opposite, saying there is a difference. For example, "The average visits are not 50."

Key Terms of Hypothesis Testing

To understand the Hypothesis testing firstly we need to understand the key terms which are given below:

- **Significance Level (α):** How sure we want to be before saying the claim is false. Usually, we choose 0.05 (5%).
- **p-value:** The chance of seeing the data if the null hypothesis is true. If this is less than α, we say the claim is probably false.
- **Test Statistic:** A number that helps us decide if the data supports or rejects the claim.
- **Critical Value:** The cutoff point to compare with the test statistic.
- **Degrees of freedom:** A number that depends on the data size and helps find the critical value.



What are Type 1 and Type 2 errors in Hypothesis Testing?

In hypothesis testing Type I and Type II errors are two possible errors that can happen when we are finding conclusions about a population based on a sample of data. These errors are associated with the decisions we made regarding the null hypothesis and the alternative hypothesis.

- **Type I error:** When we reject the null hypothesis although that hypothesis was true. Type I error is denoted by alpha(α).
- **Type II errors:** When we fail to reject the null hypothesis even though it is false, Type II errors are denoted by beta(β).

	Null Hypothesis is True	**Null Hypothesis is False**
Fail to Reject Null Hypothesis	Correct Decision	Type II Error (False Negative)
Alternative Hypothesis is True (Reject)	Type I Error (False Positive)	Correct Decision